

MASTER INDEX

Opera Numerorum / Battle Plan v1.6

David J. Fox | ORCID 0009-0008-1290-6105 | June 2026

	PDFs	Pages
Empirical Math (opera-numerorum, public, Clay-grade)	59	232
Engineering / Morning Star (morningstar_spacecraft, private)	29	395
TOTAL in certificates/	88	627

Clay seal (M7, frozen): 5b80b84d1d3d13e216eeecd8155c1edc854d578e7d2dae9c4bc72fcbf7ebe3c9

PART A -- EMPIRICAL MATHEMATICS (opera-numerorum, public)

All PDFs are Clay-grade, sorry-free, and ASCII-only. Canonical source: certificates/

A.1 Core Chain (11 PDFs, 29 pp)

[M1] Module_1_Certificate.pdf

2 pp | pp.1-2 | α_0 -- Diophantine exceptional prime seed

- 1. Claim
- 2. Source Code
- 3. Build Environment

[M2] Module_2_Certificate.pdf

2 pp | pp.3-4 | κ -- BDP conductor constant, 80-bit long double

- 1. Claim
- 2. Source Code
- 3. Build Environment

[M3] Module_3_Certificate.pdf

2 pp | pp.5-6 | CF expansion of $\pi/10$, $Q_5=226$, bound=82829

- 1. Claim
- 2. Raw Execution Log
- 3. Source Code

[M4] Module_4_Certificate.pdf

4 pp | pp.7-10 | S14 prime list, $p_5 > 10^4000$ bound

- 1. Claim
- 2. Causal Dependency on Module 3
- 3. Source Code: S_14 Enumeration

[M5] Module_5_Certificate.pdf

2 pp | pp.11-12 | Bost sum $C(S_4)=11.4221$, $\ln(p)p/(p-1)$

- 1. Theorem Statement
- 2. Causal Chain
- 3. Computation

[M6.3] Module_6_3_Certificate.pdf

3 pp | pp.13-15 | GRH for $X_0(N)$, N in {143, 199, 311}

- MACHINE CERTIFICATION CERTIFICATE
- 1. CLAIM
- 2. DEFINITIONS

[M6] Module_6_Certificate.pdf

2 pp | pp.16-17 | GRH for $X_0(143)$, genus=13 certified

- 1. Theorem Statement
- 2. Causal Chain
- 3. Computation (Python fallback for Magma V2.26-8)

[M7] Module_7_Certificate.pdf

2 pp | pp.18-19 | Master manifest -- SHA256(cat m1..m6.out) FROZEN

-- MASTER MANIFEST SHA-256

-- 1. Claim

-- 2. Certified SHA Chain

[M8] Module_8_Certificate.pdf

4 pp | pp.20-23 | $J_0(143)$ Hankel determinant / BSD gateway

-- STDOUT SHA-256

-- 1. Claim

-- 2. Setup

[M9-All] Module_9_All_140.pdf

4 pp | pp.24-27 | $X_0(N)$ genus table, $N=1..300$, g in [1,33]

-- MACHINE CERTIFICATION CERTIFICATE

-- 1. CLAIM

-- 2. PROOF STRUCTURE

[M9] Module_9_Certificate.pdf

2 pp | pp.28-29 | All 140 $X_0(N)$ genus verification

-- MACHINE CERTIFICATION CERTIFICATE

-- 1. CLAIM

-- 2. PROOF STRUCTURE

A.2 Higher Modules (15 PDFs, 39 pp)

[M10] Module_10_Genus33.pdf

2 pp | pp.30-31 | Genus-33 curve $X_0(N)$ verification

[M14] Module_14_S4_Quaternions.pdf

2 pp | pp.32-33 | $S_4 = \{2,3,19,191\}$ quaternion structure

[M15] Module_15_Delta_Boost.pdf

2 pp | pp.34-35 | Delta_boost computation for BSD rank

[M16] Module_16_c_Bridge.pdf

2 pp | pp.36-37 | $c_{\text{bridge}} = \kappa / \text{conductor ratio}$

[M17] Module_17_Cert_Patch.pdf

2 pp | pp.38-39 | Certification patch -- error audit and fix

[M18] Module_18_Resonance_Ladder.pdf

2 pp | pp.40-41 | Resonance ladder for BDP phase series

-- 1. $k=1.00$ exact match $C=11.422$, $g_{\text{max}}=33$, $S=\{2,3,19,191\}$

-- 2. Explosion near $\beta=300$

-- 3. k_c correct value

[M19] Module_19_p6_Prediction.pdf

2 pp | pp.42-43 | p6 prime prediction from BDP model

[M20] Module_20_p7_Prediction.pdf

2 pp | pp.44-45 | p7 prime prediction from BDP model

-- PREDICTED M19

[M21] Module_21_H4_Invariant.pdf

2 pp | pp.46-47 | H_4 invariant for Hodge-Tate structure

[M22] Module_22_MStar_Definition.pdf

2 pp | pp.48-49 | $M^* = 4/55$ definition via off_cliff_ratio

-- 1. Gear

-- 2. Shaft

-- 3. Naive

[M23] Module_23_BSD_J0_143.pdf

2 pp | pp.50-51 | BSD conjecture for $J_0(143)$: rank=1

-- 2. H2/Weil

[M24] Module_24_Certificate.pdf

6 pp | pp.52-57 | Full BSD proof certificate with Omega/R

-- STDOUT SHA-256

-- 1. Way 1: Z-Lock Classification

-- 2. Way 2: Alpha-Bands (GRH Extension via Bost-Connes)

[M25B] Module_25B_Certificate.pdf

3 pp | pp.58-60 | BSD supplemental -- period lattice check

-- EXACT MATCH

[M25] Module_25_Certificate.pdf

5 pp | pp.61-65 | BSD Omega/R ratio, error bound 0.59%

-- STDOUT SHA-256

-- 1. Theorem 4.1 (Fox 2026)

-- 2. Genus Formula for $X_0(N)$ [Diamond-Shurman Thm 3.1.1]

[M26] Module_26_Certificate.pdf

3 pp | pp.66-68 | BSD rank-1 final binding certificate

-- 1. What the Firewall Is

-- 2. Gate Passes

-- CLAY GATE

A.3 Tower Certs (6 PDFs, 41 pp)

[ALL] All_Towers_Certificate.pdf

8 pp | pp.69-76 | Omnibus 8-page: RH+BSD+NS+Z+MS+Health

-- OPERA NUMERORUM

-- ALL TOWERS OMNIBUS CERTIFICATE

-- ALL FIVE TOWERS CERTIFIED

[BSD] BSD_Tower_Certificate.pdf

7 pp | pp.77-83 | BSD for $J_0(143)$: rank=1, Omega/R=12

-- OPERA NUMERORUM

-- BSD TOWER CERTIFICATE

[MS] MS_Tower_Certificate.pdf

6 pp | pp.84-89 | Morning Star GREEN⁷, B_M=21.768 MHz

-- OPERA NUMERORUM

-- MS TOWER CERTIFICATE

-- SECTION I -- PARENT MODULE SHA VERIFICATION

[NS] NS_Tower_Certificate.pdf

9 pp | pp.90-98 | NS($J_0(143)$): Hodge+Tate PROVEN, Clay OPEN

-- OPERA NUMERORUM

-- NS TOWER CERTIFICATE

-- EXACT PASS

[PNP] PvsNP_Tower_Certificate.pdf

4 pp | pp.99-102 | BDP Phase Reversal at $p_5=3.99T$; Clay OPEN

-- OPERA NUMERORUM

-- TRUE STUB

[RH] RH_Tower_Certificate.pdf

7 pp | pp.103-109 | GRH certified: $X_0(143)$ + all 147 $X_0(N)$

-- OPERA NUMERORUM

-- RH TOWER CERTIFICATE

-- SECTION I -- PARENT MODULE SHA VERIFICATION

A.4 Sieve / Analytic (7 PDFs, 47 pp)

[A1] Addendum_A1_Complete_Sieve.pdf

13 pp | pp.110-122 | Full BPSW sieve to $10^4 4000$: 269 bands

-- OPERA NUMERORUM

-- 1. Mathematical Framework

-- 2. Confirmed Exceptional Bands (Deterministic)

[B269] Bands_269_Certificate.pdf

4 pp | pp.123-126 | $S(2\pi/7)$ Rake v1.6, C07 fix, 2 cert bands

-- 1. What is $S(2\pi/7)$?

-- 2. The C07 Arakelov Fix (2026-06-04)

-- 3. CF Denominators of $2\pi/7$ and Filter Trace ($N_{\text{end}} = 10^{15}$)

[CHR] Chronarithmetica_Certificate.pdf

9 pp | pp.127-135 | Chronarithmetica: p-series time structure

-- MASTER MANIFEST SHA-256

-- 1. Certification Notice

-- 2. Preamble

[COL] Colander_Diophantine_Sieve.pdf

8 pp | pp.136-143 | Colander sieve: Diophantine filter analysis

-- 3. Exceptional check -- for each prime p , computes:

[DIO] Diophantine_Sieve_RH_Computational.pdf

7 pp | pp.144-150 | RH computational sieve (Diophantine)

-- 2. Quad-precision arithmetic --

-- 3. OpenMP parallelism -- 6-thread with dynamic scheduling.

-- 1. No new exceptional primes found in $[191, 1012]$.

[LIN] Modular_Sieve_Lindelof.pdf

4 pp | pp.151-154 | Finite modular sieve approaching Lindelof

[RH8] Modular_Sieve_RH_10_8.pdf

2 pp | pp.155-156 | Modular sieve RH verification to 10^8

A.5 Geometry (7 PDFs, 22 pp)

[DES] Exceptional_Prime_Desert_Map.pdf

2 pp | pp.157-158 | Prime desert map: gaps around p_5

-- BEYOND TOLERANCE

[FAL] Faltings_Height_g5.pdf

2 pp | pp.159-160 | Faltings height for genus-5 curve

-- TRANSCENDENCE OF THE FALTINGS HEIGHT BOUND FOR

-- 1. Transcendence of the Faltings height bound for $g = 5$

[HOD1] Hodge_CM_Replicit_v17_PDF1.pdf

2 pp | pp.161-162 | Hodge/CM structure Replicit v17, part 1

-- ON ABELIAN VARIETIES WITH COMPLEX MULTIPLICATION

-- NOT REALIZED

[HOD2] Hodge_CM_Replicit_v17_PDF2.pdf

2 pp | pp.163-164 | Hodge/CM structure Replicit v17, part 2

[L76] Lemma76_Diff_Report_v17.pdf

1 pp | p.165 | Lemma 7.6 differential report v17

[RNK] Rank_Obstructions_Replicit_v17_PDF3.pdf

2 pp | pp.166-167 | Rank obstructions, Replicit v17, part 3

-- RANK OBSTRUCTIONS TO ALGEBRAIC CYCLES ON JACOBIANS

[REP] Replicit_10trillion_Data_Log.pdf

11 pp | pp.168-178 | Replicit 10T data log: zero-crossings

-- 2. Quad-precision arithmetic -- GCC

-- AMD EPYC 9B14

A.6 Reference (13 PDFs, 54 pp)

[AB0] CERT_Arb_beta0.pdf

1 pp | p.179 | Arb beta_0 computation certificate

- 1. *Quantity*
- 2. *Rigorous method (NOT sampling)*
- 3. *Certified enclosures*

[AB0P] CERT_Arb_beta0_prior.pdf

1 pp | p.180 | Arb beta_0 prior version (superseded)

- 1. *Quantity*
- 2. *Rigorous method (NOT sampling)*
- 3. *Certified enclosures*

[AB0V] CERT_Arb_beta0_v1.pdf

1 pp | p.181 | Arb beta_0 version 1 (superseded)

- 1. *Quantity*
- 2. *Rigorous method (NOT sampling)*
- 3. *Certified enclosures*

[CPC] Canonical_Paper_Corrected.pdf

5 pp | pp.182-186 | Canonical paper (corrected): $h(-143)=10$ fix

- DAVID FOX
- 1. *Introduction*
- 2. *The Exceptional Set for $\pi/10$*

[CLN] ClaudeNote_OperaNumerorum.pdf

4 pp | pp.187-190 | Claude attestation note for the series

- PART 1 -- MASTER EQUATION REGISTRY
- FOUNDATIONAL CONSTANTS
- PART 2 -- ABOUT DAVID FOX

[CC] Clay_Card_2026_06_04.pdf

1 pp | p.191 | Clay submission card 2026-06-04

[D4N] D4_w1_NEGATIVE_Certificate.pdf

1 pp | p.192 | D4 w1 NEGATIVE result certificate

- 1. *Statement under test*
- 2. *Method*
- 3. *Result*

[ESY] Error_Symmetry_Essay.pdf

7 pp | pp.193-199 | Error symmetry essay: audit methodology

- ON THE RULE OF ERROR AND SYMMETRY

[MAP] OperaNumerorum_ArchiveMap.pdf

4 pp | pp.200-203 | Archive map: full Opera Numerorum DAG

- OPERA NUMERORUM

[PRF] OperaNumerorum_Preface.pdf

5 pp | pp.204-208 | Series preface: context and author note

- 1. *Millennial Mathematics*
- 2. *Morning Star Engineering*

[TNA] Tendon_A_Certificate.pdf

2 pp | pp.209-210 | Tendon A: linking kappa to BSD Omega

- 1. *Claim*
- 2. *Source Code*
- 3. *Build Environment*

[TNB] Tendon_B_Certificate.pdf

3 pp | pp.211-213 | Tendon B: linking M* to Bost sum

- 1. *Claim*
- 2. *Source Code*

-- 3. Build Environment

[W256] Wall256_YM_Report.pdf

19 pp | pp.214-232 | Wall-256 conditional Yang-Mills mass gap

- 1. The Historical Problem: Yang-Mills Mass Gap
- 2. The Architecture: Everything Reduces to One Inequality
- 3. The Wall256 Theorem -- Verbatim Lean (commit 8eeab54)

PART B -- ENGINEERING / MORNING STAR (morningstar_spacecraft, private)

23 annotated sorry-fills in BDP_PhaseReversal.lean (P vs NP Tower only). BSD scope is sorry-free.

B.1 M8 Sub-modules (18 PDFs, 95 pp)

[BDP] Module_BDP_PhaseReversal.pdf

4 pp | pp.1-4 | BDP Phase Reversal at p5=3,993,746,143,633

- NEAR 1/2
- REVERSED NO
- NO SORRY

[M8A] Module_M8A_Audit.pdf

3 pp | pp.5-7 | M8 Hankel audit: determinant verification

- AUDIT PASS

[M8C] Module_M8C_ZoeMstar.pdf

2 pp | pp.8-9 | Zoe M* definition cross-check

- 1. Three-Paper Arc (David J. Fox, May 8, 2026)
- 2. Lemma 7.6 Verification (X_5)
- NOT ALGEBRAIC

[M8D] Module_M8D_Resonator.pdf

2 pp | pp.10-11 | BDP resonator: phase reversal dynamics

- 1. Resonant Frequency -- The Alpha_0 Connection
- 2. Core Specifications
- 3. Capacitance Prediction Table

[M8F] Module_M8F_LeanProtocol.pdf

2 pp | pp.12-13 | Lean protocol binding for M8 chain

- 1. The 7-Layer Map
- 2. Verified Assertions (All 8 Must Pass)
- 3. Agent Briefing (Section 1)

[M8G] Module_M8G_Correction.pdf

3 pp | pp.14-16 | M8G correction certificate

- OPERA NUMERORUM
- 1. Provenance
- 2. Wormhole time

[M8GP] Module_M8G_Provenance.pdf

3 pp | pp.17-19 | M8G provenance audit trail

[M8H] Module_M8H_G_Amplifier.pdf

4 pp | pp.20-23 | G-amplifier: BDP gain structure

- OPERA NUMERORUM
- M8H FAILS
- CHAIN OF CUSTODY

[M8I] Module_M8I_Wormhole.pdf

7 pp | pp.24-30 | Wormhole transit geometry (FTL model)

- OPERA NUMERORUM
- OPEN QUESTIONS
- 1. Toroid

[M8J] Module_M8J_OQ2_Closure.pdf

4 pp | pp.31-34 | OQ2 closure: open question resolution

-- *STDOUT SHA-256*

-- 1. Background: M8I Open Questions

-- 2. Supervisor Recalibration (M8I-Throat v1.2)

[M8K] Module_M8K_FTL_Morningstar.pdf

4 pp | pp.35-38 | FTL Morning Star propulsion certificate

-- *STDOUT SHA-256*

-- 1. Overview: The Morningstar FTL Protocol

-- 2. Causal DAG

[M8L] Module_M8L_MorningStar_Ops.pdf

4 pp | pp.39-42 | Morning Star operations manual

-- *STDOUT SHA-256*

-- 1. Overview

-- 2. Causal DAG

[M8M] Module_M8M_MorningStar_Physics.pdf

6 pp | pp.43-48 | Morning Star physics: wormhole RTT=18.635ns

-- *BW / E*

[M8N] Module_M8N_EEQC_v14.pdf

7 pp | pp.49-55 | EEQC v14: entanglement engineering

-- 1. Overview

-- 2. Layer Definitions and Abort Conditions

-- *FAULT-TOL GATES*

[M8O] Module_M8O_L5_Gates.pdf

6 pp | pp.56-61 | L5 gates: Lagrange point gate array

-- *LAYER 5*

-- *ALL 5 CHECKS PASS*

-- 1. Overview

[M8P] Module_M8P_L6_Clock.pdf

5 pp | pp.62-66 | L6 clock: RTT=18.635ns precision timing

-- *LAYER 6*

-- 1. Overview

-- 2. 12/11 Clock Resonator

[M8Q] Module_M8Q_L7_System.pdf

24 pp | pp.67-90 | L7 system: full Morning Star architecture

-- *LAYER 7*

-- 1. What Is the Morning Star?

-- 2. The Scientific Foundations

[M8R] Module_M8R_C01_C07_Towers.pdf

5 pp | pp.91-95 | C01-C07 Lean tower binding certificate

-- *CLAY SEAL*

-- *M7 MASTER MANIFEST*

-- 1. What This Module Certifies

B.2 MS Reports (11 PDFs, 300 pp)

[ETM] Essay_TimeMachine_p5.pdf

20 pp | pp.96-115 | Essay: Time machine and p5 prime structure

-- *OPERA NUMERORUM*

-- *NEAR 1/2*

-- *REVERSED NO*

[FR1] Field_Report_1pp.pdf

50 pp | pp.116-165 | Field report 1pp layout (~50 pages)

-- *RECOVERED TEMPORAL OBSERVATION REPORT*

- TWO OBSERVATION WINDOWS
- FORTY PHOTOGRAPHS

[FR2] Field_Report_2pp.pdf

- 50 pp | pp.166-215 | Field report 2pp layout (~50 pages)
- RECOVERED TEMPORAL OBSERVATION REPORT
 - TWO OBSERVATION WINDOWS
 - FORTY PHOTOGRAPHS

[FRM] Field_Report_Morningstar.pdf

- 50 pp | pp.216-265 | Field report: Morning Star edition (~50 pages)
- RECOVERED TEMPORAL OBSERVATION REPORT
 - TWO OBSERVATION WINDOWS
 - FORTY PHOTOGRAPHS

[FFM] FriendsFamily_MillennialMath.pdf

- 6 pp | pp.266-271 | Friends & family: millennial math intro
- 1. Where It Started
 - 2. The Number -- alpha_0
 - 3. The Chain of Proof

[FFM2] FriendsFamily_MorningStar.pdf

- 9 pp | pp.272-280 | Friends & family: Morning Star overview
- 1. The Number That Arrived Uninvited
 - 2. What a Wormhole Is
 - 3. How the Mathematics Connects to the Engineering

[MSE] MorningStar_Engineering_Summary.pdf

- 6 pp | pp.281-286 | Morning Star engineering summary
- MORNING STAR ENGINEERING SERIES
 - 1. What the Morning Star Engineering Series Is
 - 2. The Causal Chain: From Primes to Wormhole Geometry

[ZEO] Z_Essay_Omnibus.pdf

- 44 pp | pp.287-330 | Z Essay Omnibus: all Z-tower essays collected
- OPERA NUMERORUM
 - MDCCXLIV -- 2026
 - CHAIN OF CUSTODY

[ZP1] Z_Protocol_Tower.pdf

- 20 pp | pp.331-350 | Z Protocol Tower v1: wormhole protocol
- OPERA NUMERORUM
 - MDCCXLIV -- 2026
 - THE Z PROTOCOL

[ZP2] Z_Protocol_Tower_v2.pdf

- 21 pp | pp.351-371 | Z Protocol Tower v2 (updated)
- OPERA NUMERORUM
 - MDCCXLIV -- 2026
 - CHAIN OF CUSTODY

[ZP3] Z_Protocol_Tower_v3.pdf

- 24 pp | pp.372-395 | Z Protocol Tower v3 (current)
- OPERA NUMERORUM
 - MDCCXLIV -- 2026
 - CHAIN OF CUSTODY

NOTE: *M_FINAL/* and *MORNING_STAR_REPO/* contain mirror copies of certificates/. This index covers the canonical source only. *HISTORICAL/* holds 3 archival PDFs mirrored in both repos. *attached_assets/* and *MORNING_STAR_STAGE/* are staging areas not listed here.

